

MICROSOFT POWER AUTOMATE

Hands-On Lab Exercise Workbook

9 Practical Labs | Beginner to Intermediate

2-Day Corporate Training Programme

Participant Name: _____

Department: _____ Date: _____

Trainer: Fakhrul Syahmi | mufasyahcons@gmail.com

How to Use This Workbook

This lab workbook is your hands-on companion for the Microsoft Power Automate: Workflow Automation for Workplace Productivity training programme. Each lab corresponds to a real business scenario you will automate step by step during the training session.

Feature	Description
9 Practical Labs	Cover email automation, approvals, Teams notifications, SharePoint, Excel, AI Builder and more
Scenario-Based	Each lab uses a realistic business context you can relate to
Step-by-Step	Instructions guide you through every click and configuration
Validation Checklists	Checkbox lists confirm your flow is working correctly
Reflection Space	Write-in sections to record your observations and learnings
Troubleshooting Tips	Common errors and solutions for each lab

IMPORTANT: Before starting any lab, make sure you are signed into make.powerautomate.com with your Microsoft 365 work account. Keep this URL open in your browser throughout the training.

Lab Overview & Day Mapping

Lab	Day
Lab 1: Auto-Save Email Attachments to OneDrive	Day 1 — Morning
Lab 2: Leave Approval Automation	Day 1 — Afternoon
Lab 3: Teams Notification Flow	Day 1 — Afternoon
Lab 4: SharePoint Document Approval	Day 2 — Morning
Lab 5: Excel Data Update (Scheduled)	Day 2 — Morning
Lab 6: Daily Reminder Flow	Day 2 — Afternoon
Lab 7: Invoice Approval (Multi-Level)	Day 2 — Afternoon
Lab 8: Employee Onboarding Workflow	Day 2 — Afternoon
Lab 9: AI Email Classification Flow	Day 2 — Closing

LAB 1 — Auto-Save Email Attachments to OneDrive

Business Scenario

Siti works in the Finance department at Petronas Sdn Bhd. Every day, vendors send invoice PDFs to her email. She spends 30 minutes each morning manually downloading each attachment and organising them into folders. After this lab, the entire process will happen automatically — the moment an email with an attachment arrives, it is saved to the correct OneDrive folder instantly.

Component	Configuration
Trigger	When a new email arrives (Outlook)
Filter	Subject contains 'INVOICE'
Condition	Has Attachments = Yes
Action 1	Loop through all attachments (Apply to Each)
Action 2	Create file in OneDrive /Finance/Invoices
Action 3	Send confirmation email to Siti
Connector	Outlook + OneDrive for Business
Estimated Build Time	20–25 minutes

Pre-Lab Setup

- Log in to make.powerautomate.com
- Log in to onedrive.com and create folder: Finance > Invoices
- Have your Outlook inbox accessible
- Prepare a test email with a PDF or Word attachment (you will send this to test)

Step-by-Step Instructions

STEP 1: Create a New Automated Cloud Flow

1. In Power Automate, click Create in the left navigation
2. Select Automated cloud flow
3. In the flow name field, type: Finance-AutoSave-InvoiceAttachments
4. In the trigger search box, type: new email
5. Select When a new email arrives (V3) — from Outlook connector
6. Click Create

STEP 2: Configure the Trigger

7. In the trigger settings, click Show advanced options
8. Set Subject Filter to: INVOICE
9. Set Include Attachments to: Yes
10. Set Only with Attachments to: Yes

11. Leave all other settings as default

WHY: The Subject Filter ensures your flow only fires for emails with INVOICE in the subject line. This prevents the flow from running for every email you receive.

STEP 3: Add Apply to Each Loop

12. Click + New Step
13. In the search bar, type: Apply to each
14. Select Apply to each (Control)
15. In the Select an output from previous steps field, click the dynamic content panel
16. Search for and select: Attachments (from the trigger)

STEP 4: Add Create File Action Inside the Loop

17. Inside the Apply to Each block, click Add an action
18. Search for: Create file
19. Select Create file from OneDrive for Business
20. Folder Path: click the folder icon and navigate to /Finance/Invoices
21. File Name: click inside the field, then select Attachment Name from dynamic content
22. File Content: click inside the field, then select Attachment Content from dynamic content

STEP 5: Add Confirmation Email (After the Loop)

23. Click below the Apply to Each block and click + New Step
24. Search for: Send an email
25. Select Send an email (V2) from Office 365 Outlook
26. To: enter your own email address
27. Subject: type — Invoice Attachment Saved Automatically
28. Body: compose — New invoice email received from [insert From dynamic content]. Attachment has been saved to /Finance/Invoices on OneDrive.

STEP 6: Save and Test

29. Click Save (top right)
30. Send yourself a test email with INVOICE in the subject and a PDF attachment
31. Wait 1–2 minutes for the flow to trigger
32. Check your OneDrive Finance > Invoices folder for the saved file
33. Check your inbox for the confirmation email
34. Go to My Flows > click your flow > check 28 day run history — it should show Succeeded

Validation Checklist

Done?	Checklist Item
☐	Flow is saved and shows as On status
☐	Trigger fires only for emails with INVOICE in the subject line

☐	All email attachments appear in /Finance/Invoices folder on OneDrive
☐	File names match the original attachment names
☐	Confirmation email received with sender details in the body
☐	Run History shows green checkmark (Succeeded) for all steps
☐	Apply to Each loop shows correct number of iterations

Troubleshooting Guide

Problem	Solution
Flow did not fire after sending email	Check Subject Filter — it must match exactly (case-insensitive). Also check trigger settings: Include Attachments = Yes.
File saved but wrong name	Check that Attachment Name dynamic content is selected (not file name from another step).
OneDrive folder not found	Create the /Finance/Invoices folder manually in OneDrive before running the flow.
Run History shows failed	Click the failed run, expand each step to find the red X, read the error message carefully.

My Notes & Observations

What did the flow do when you tested it?

Any errors you encountered and how you fixed them:

How could you use this flow in your actual job?

LAB 2 — Leave Approval Automation

Business Scenario

TechCorp Sdn Bhd's HR team processes 80+ leave requests per month manually. Each request involves reading an email, forwarding to the manager, waiting for a reply, updating the Excel tracker, and notifying the employee. This lab automates the entire process using Microsoft Forms, Power Automate Approvals, and Excel Online.

Component	Configuration
Trigger	When a new response is submitted (Microsoft Forms)
Action 1	Get response details
Action 2	Start and wait for an approval
Action 3	Condition: Outcome = Approve
Action 4 (Yes)	Send approval email to employee
Action 4 (No)	Send rejection email to employee
Action 5 (Both)	Add row to Excel leave tracker
Connector	Forms + Approvals + Outlook + Excel Online
Estimated Build Time	30–35 minutes

Pre-Lab Setup

Step A: Create the Microsoft Form

35. Open forms.microsoft.com
36. Click + New Form
37. Title: Employee Leave Request
38. Add fields:
39. Question 1: Employee Name (Text — Required)
40. Question 2: Department (Choice: HR, Finance, IT, Marketing, Operations, Sales, Admin)
41. Question 3: Leave Type (Choice: Annual Leave, Medical Leave, Emergency Leave, Maternity Leave)
42. Question 4: Start Date (Date — Required)
43. Question 5: End Date (Date — Required)
44. Question 6: Number of Days (Number — Required)
45. Question 7: Reason (Text, Long Answer — Required)
46. Click Share and copy the form link for testing

Step B: Create the Excel Leave Tracker

47. Open OneDrive and create a new Excel file: Leave_Tracker_2026.xlsx
48. In Sheet1, create a table with these column headers (format as Table):

49. Request ID | Employee Name | Department | Leave Type | Start Date | End Date | Days | Status | Manager Decision | Date Processed
50. Name the table: LeaveData (Table Design > Table Name)
51. Save and close the file

Building the Flow

STEP 1: Create the Flow

52. Go to Power Automate > Create > Automated cloud flow
53. Name: HR-LeaveRequest-ApprovalFlow
54. Trigger: When a new response is submitted (Microsoft Forms)
55. Form ID: select Employee Leave Request from dropdown
56. Click Create

STEP 2: Get Response Details

57. Click + New Step
58. Search: Get response details
59. Select Get response details (Microsoft Forms)
60. Form ID: Employee Leave Request
61. Response ID: select Response Id from dynamic content (trigger)

STEP 3: Start Approval

62. Click + New Step > Search: Start and wait for an approval
63. Approval Type: Approve/Reject — First to Respond
64. Title: Leave Request — then insert Employee Name from dynamic content
65. Assigned to: enter your trainer's email OR your manager's email
66. Details: compose a clear message using dynamic content fields:
67. Employee: [Employee Name]
68. Department: [Department]
69. Leave Type: [Leave Type]
70. Start Date: [Start Date]
71. End Date: [End Date]
72. Days Requested: [Number of Days]
73. Reason: [Reason]

STEP 4: Add Condition

74. Click + New Step > Search: Condition
75. Left field: select Outcome from dynamic content (Start and wait for an approval)
76. Operator: is equal to
77. Right field: type Approve

STEP 5: Configure YES Branch (Approved)

78. In If Yes, click Add an action

- 79. Search: Send an email (V2)
- 80. To: use Employee Email OR your own email for testing
- 81. Subject: Your Leave Request Has Been APPROVED
- 82. Body: compose congratulations message with leave details from dynamic content

STEP 6: Configure NO Branch (Rejected)

- 83. In If No, click Add an action > Send an email (V2)
- 84. Subject: Your Leave Request Status Update
- 85. Body: inform that request was not approved, include manager comments from Approver's Response Comments

STEP 7: Add Excel Row (Both Branches)

- 86. Add action in BOTH Yes and No branches:
- 87. Search: Add a row into a table
- 88. Select Add a row into a table (Excel Online — Business)
- 89. Location: OneDrive for Business
- 90. Library: OneDrive
- 91. File: Leave_Tracker_2026.xlsx
- 92. Table: LeaveData
- 93. Map all columns to dynamic content values
- 94. Status column: type Approved (in Yes branch) or Rejected (in No branch)

Test Your Flow

- 95. Save the flow
- 96. Open the Microsoft Form link and submit a test leave request
- 97. Check the approver's email or Teams for the approval request
- 98. Approve or reject the request
- 99. Verify the employee notification email arrives
- 100. Check the Excel tracker for the new row

Validation Checklist

Done?	Checklist Item
☐	Microsoft Form has all 7 required fields
☐	Flow triggers when form is submitted
☐	Approval email arrives with all leave details
☐	Approver can approve/reject from email or Teams
☐	Employee receives correct email based on approval decision
☐	Excel tracker shows new row with correct data

☐	Run History shows Succeeded for all steps
☐	Flow handles both Approved and Rejected paths correctly

Troubleshooting Guide

Problem	Solution
Approval email not received	Check the Assigned To field — email must be correct and an active Microsoft 365 user. Also check Junk/Clutter folder.
Dynamic content missing after Get response details	Ensure the Form ID in Get response details matches the Form ID in the trigger exactly.
Excel row not added	Verify the Excel file is in OneDrive (not SharePoint), table is named correctly, and column names match exactly.
Outcome condition not matching	Approval outcomes are case-sensitive. Use 'Approve' not 'approve' or 'Approved'.

My Notes & Observations

Where did the approval request appear for the approver?

What dynamic content did you use in the approval Details field?

How would you modify this flow for your department's leave process?

LAB 3 — Teams Notification Flow

Business Scenario

The Project Management team at DataTech Corp uses a SharePoint list called Project Tasks to track all ongoing work. Currently, team members have to manually check the list to know when a new task is assigned. This lab automates the notification — every time a new task is added to SharePoint, an instant Teams message is posted to the team channel and the assigned person gets an email.

Component	Configuration
Trigger	When an item is created (SharePoint)
Action 1	Post a message in a Teams channel
Action 2	Send email to the person assigned to the task
Connector	SharePoint + Microsoft Teams + Outlook
Estimated Build Time	15–20 minutes

Pre-Lab Setup

101. Open SharePoint (your trainer will provide the site URL)
102. Create a new List named: Project Tasks
103. Add columns: Task Name (Title — already exists), Assigned To (Person), Priority (Choice: High/Medium/Low), Due Date (Date), Status (Choice: Not Started/In Progress/Completed), Notes (Multiple lines of text)
104. Add 1 sample item manually to verify the list works

Building the Flow

STEP 1: Create the Flow

105. Power Automate > Create > Automated cloud flow
106. Name: PM-NewTask-TeamsNotification
107. Trigger: When an item is created (SharePoint)
108. Site Address: select your SharePoint site
109. List Name: Project Tasks
110. Click Create

STEP 2: Post to Teams Channel

111. Click + New Step
112. Search: Post a message in a chat or channel
113. Post as: Flow bot
114. Post in: Channel
115. Team: select your training team
116. Channel: select General (or your project channel)

- 117. Message: compose using dynamic content:
- 118. NEW TASK CREATED
- 119. Task: [Title]
- 120. Assigned To: [Assigned To Display Name]
- 121. Priority: [Priority]
- 122. Due Date: [Due Date]
- 123. Status: [Status]

STEP 3: Send Email to Assigned Person

- 124. Click + New Step
- 125. Search: Send an email (V2)
- 126. To: select Assigned To Email from dynamic content
- 127. Subject: New Task Assigned to You — then add Title dynamic content
- 128. Body: compose a friendly message with task details
- 129. Include a note: Please log in to SharePoint to view the full task details and update your status.

Test Your Flow

- 130. Save the flow
- 131. Add a new item to the Project Tasks SharePoint list
- 132. Wait 1 minute and check the Teams channel for the notification
- 133. Check the assigned person's email (use your own email for testing)
- 134. Verify Run History shows Succeeded

Validation Checklist

Done?	Checklist Item
☐	Flow triggers when new item is added to SharePoint list
☐	Teams message appears in correct channel within 1-2 minutes
☐	Teams message includes Task Name, Assigned Person, Priority, and Due Date
☐	Email sent to assigned person with task details
☐	Run History shows all steps succeeded
☐	No duplicate messages sent

My Notes & Observations

What appeared in Teams when you added the SharePoint item?

What other SharePoint triggers could you use in your work?

LAB 4 — SharePoint Document Approval Process

Business Scenario

GreenBuild Bhd's procurement team uploads vendor contracts to a SharePoint document library. Currently, a manager must be manually notified to review each document. This lab creates an automated approval flow: when any document is uploaded to the Pending Approval library, the manager receives an approval request. Approved documents move to the Approved folder; rejected ones are moved to Rejected with a reason.

Component	Configuration
Trigger	When a file is created (SharePoint document library)
Action 1	Start and wait for an approval with link to document
Action 2	Condition: Outcome = Approve
Yes Branch	Copy file to Approved Documents library
Yes Branch	Update SharePoint item status column
No Branch	Move file to Rejected Documents library
Both Branches	Email uploader with decision + reason
Estimated Build Time	30–35 minutes

Pre-Lab Setup

135. In SharePoint, create 3 document libraries: Pending Approval, Approved Documents, Rejected Documents
136. Add a Status column to Pending Approval library (Choice: Pending Review, Approved, Rejected)
137. Have a sample Word or PDF document ready to upload for testing

Building the Flow

STEP 1: Create the Flow

138. Power Automate > Create > Automated cloud flow
139. Name: Procurement-DocApproval-Flow
140. Trigger: When a file is created (properties only) — SharePoint
141. Site Address: your SharePoint site
142. Library Name: Pending Approval
143. Click Create

STEP 2: Start Approval

- 144. + New Step > Start and wait for an approval
- 145. Type: Approve/Reject — First to Respond
- 146. Title: Document Review Required — [File name] (dynamic content)
- 147. Assigned to: enter manager email
- 148. Details: include file name, uploaded by, date
- 149. Item link: insert Link to item from dynamic content
- 150. Item link description: Click here to view the document

STEP 3: Add Condition and Branches

- 151. + New Step > Condition
- 152. Outcome is equal to Approve
- 153. YES Branch: Copy file → Approved Documents library
- 154. YES Branch: Update file properties — Status = Approved
- 155. YES Branch: Send email — Document approved, saved to Approved Documents
- 156. NO Branch: Copy file → Rejected Documents library
- 157. NO Branch: Update file properties — Status = Rejected
- 158. NO Branch: Send email — Document not approved, include Approver's Comments

Validation Checklist

Done?	Checklist Item
☐	Flow triggers when file is uploaded to Pending Approval library
☐	Approval email sent to manager with link to document
☐	Manager can open document from approval email link
☐	Approved documents appear in Approved Documents library
☐	Rejected documents appear in Rejected Documents library
☐	Status column updated correctly in SharePoint
☐	Uploader receives decision notification email
☐	Run History shows Succeeded

My Notes & Observations

What challenges did you face configuring the Copy file action?

How would a multi-level approval look for high-value contracts?



LAB 5 — Scheduled Excel Data Update

Business Scenario

Every Monday morning, the Operations Manager at LogisticsPro needs an up-to-date dashboard of the week's activities. Instead of manually compiling data, this scheduled flow automatically logs the weekly run date and status into an Excel tracker, then emails the team a summary report to start the week.

Component	Configuration
Trigger	Recurrence (every Monday at 8:00 AM)
Action 1	Initialize Variable: WeekStart date
Action 2	Add a row to Excel Weekly Log table
Action 3	Send email summary to operations team
Connector	Schedule + Variables + Excel Online + Outlook
Estimated Build Time	15–20 minutes

Pre-Lab Setup

159. In OneDrive, create a new Excel file: Ops_Weekly_Log.xlsx
160. In Sheet1, create a table named WeeklyLog with columns:
161. Week_Start | Run_DateTime | Department | Status | Notes | Generated_By
162. Save and close the file

Building the Flow

STEP 1: Create the Flow

163. Power Automate > Create > Scheduled cloud flow
164. Name: Ops-Weekly-ExcelUpdate
165. Starting: select next Monday
166. Repeat every: 1 Week, on Monday
167. Click Create

STEP 2: Initialize Variable

168. + New Step > Initialize variable
169. Name: WeekStartDate
170. Type: String
171. Value: click Expression tab, type: formatDateTime(utcNow(),'dd/MM/yyyy')
172. Click OK

STEP 3: Add Row to Excel

173. + New Step > Add a row into a table (Excel Online)

- 174. Location: OneDrive for Business
- 175. Library: OneDrive
- 176. File: Ops_Weekly_Log.xlsx
- 177. Table: WeeklyLog
- 178. Map columns:
- 179. Week_Start → variable WeekStartDate
- 180. Run_DateTime → expression: utcNow()
- 181. Department → Operations
- 182. Status → Active
- 183. Notes → Automated weekly entry
- 184. Generated_By → Power Automate

STEP 4: Send Summary Email

- 185. + New Step > Send an email (V2)
- 186. To: your email address
- 187. Subject: Weekly Operations Log Updated — [WeekStartDate variable]
- 188. Body: The weekly operations log has been updated automatically. Week Start: [WeekStartDate]. Please open the tracker to view this week's full report.

STEP 5: Test Immediately

- 189. Save the flow
- 190. Click Test > Manually > Run flow
- 191. Check the Excel file for the new row
- 192. Check your email for the summary notification

Validation Checklist

Done?	Checklist Item
☐	Flow is set to run every Monday at 8:00 AM
☐	Date variable shows correct formatted date (dd/MM/yyyy)
☐	New row added to Excel WeeklyLog table with all columns populated
☐	Summary email received with correct week date in subject
☐	Run History shows Succeeded when tested manually
☐	Flow status shows On (not disabled)

My Notes & Observations

What does the formatDateTime expression do?

How would you modify this to add data from SharePoint instead of static text?

LAB 6 — Daily Task Reminder Flow

Business Scenario

The Project Management team at DataTech Corp has staff who sometimes forget about pending tasks. This flow runs every morning at 9:00 AM, checks the SharePoint Project Tasks list for all Pending items, and sends reminder notifications to the relevant team members.

Component	Configuration
Trigger	Recurrence (every day at 9:00 AM)
Action 1	Get items from SharePoint (filter: Status = 'Not Started')
Action 2	Condition: is array empty?
Action 3	Apply to Each — loop through pending tasks
Action 4	Post Teams message per task
Action 5	Check due date — if within 2 days, send urgent email
Estimated Build Time	25–30 minutes

Building the Flow

STEP 1: Create Scheduled Flow

193. Power Automate > Create > Scheduled cloud flow
194. Name: PM-DailyReminder-PendingTasks
195. Repeat every: 1 Day at 9:00 AM
196. Click Create

STEP 2: Get Pending Tasks from SharePoint

197. + New Step > Get items (SharePoint)
198. Site Address: your SharePoint site
199. List Name: Project Tasks
200. Click Show advanced options
201. Filter Query: Status eq 'Not Started'
202. Top Count: 50 (to limit results)

STEP 3: Check if Any Tasks Exist

203. + New Step > Condition
204. Left field: click Expression tab, type: length(body('Get_items'))?['value']
205. Operator: is greater than
206. Right field: 0

STEP 4: In YES branch — Apply to Each

- 207. Add action: Apply to each
- 208. Select output: value (from Get items)
- 209. Inside loop: + Add an action > Post a message in Teams
- 210. Message: REMINDER — Pending Task: [Title] | Assigned: [Assigned To] | Due: [Due Date]
- 211. Inside loop: + Add an action > Condition
- 212. Check if Due Date is less than addDays(utcNow(),2)
- 213. YES: Send email — URGENT: Task due within 2 days!

STEP 5: In NO branch

- 214. Add action: Post a message in Teams
- 215. Message: Good morning! No pending tasks found today. Great work team!

Validation Checklist

Done?	Checklist Item
☐	Scheduled trigger set correctly to 9:00 AM daily
☐	SharePoint Filter Query returns only Not Started items
☐	Loop iterates through each task correctly
☐	Teams message posted for each pending task
☐	Urgent email sent for tasks due within 2 days
☐	Flow handles empty task list gracefully (NO branch)
☐	Run History shows Succeeded

My Notes & Observations

What does the length() expression check?

How would you modify this for your team's reporting needs?

LAB 7 — Multi-Level Invoice Approval

Business Scenario

Megamart Retail Bhd receives vendor invoices daily. Invoices below RM5,000 need only the Finance Manager's approval. Invoices RM5,000 and above require the Finance Manager AND the CFO to both approve. This lab implements the branching approval logic using Conditions and the Approvals connector.

Component	Configuration
Trigger	When a new response is submitted (Forms — Invoice Submission)
Action 1	Get response details
Action 2	Initialize Variable: InvoiceAmount (Integer)
Action 3	Set Variable from form response
Action 4	Condition: Amount >= 5000
YES Branch	Sequential approval: Finance Manager then CFO
NO Branch	Single approval: Finance Manager only
Both Branches	Excel tracker update + email notification
Estimated Build Time	35–40 minutes

Pre-Lab Setup

216. Create Microsoft Form: Invoice Submission
217. Fields: Vendor Name (Text), Invoice Number (Text), Invoice Amount (Number), Department (Choice), Description (Text), Invoice Date (Date)
218. Create Excel file: Invoice_Tracker.xlsx in OneDrive with table InvoiceLog
219. Columns: Invoice No | Vendor | Department | Amount | Status | Approved By | Decision Date | Notes

Building the Flow

STEP 1: Create and Configure Trigger

220. Create Automated cloud flow: Finance-InvoiceApproval-Flow
221. Trigger: When a new response is submitted — Invoice Submission form
222. Add: Get response details

STEP 2: Set Up Variable

223. + New Step > Initialize variable
224. Name: InvoiceAmount, Type: Float, Value: 0

225. + New Step > Set variable
 1. Name: InvoiceAmount
 2. Value: select Invoice Amount from Get response details dynamic content

STEP 3: Add Condition

3. + New Step > Condition
4. Left: InvoiceAmount (variable)
5. Operator: is greater than or equal to
6. Right: 5000

STEP 4: YES Branch — Dual Approval

7. Add: Start and wait for an approval
8. Type: Approve/Reject — Everyone must approve
9. Assigned to: Finance Manager email; CFO email (comma separated)
10. Title: HIGH-VALUE Invoice Approval Required — [Vendor Name] — RM[InvoiceAmount]

STEP 5: NO Branch — Single Approval

11. Add: Start and wait for an approval
12. Type: Approve/Reject — First to Respond
13. Assigned to: Finance Manager email only
14. Title: Invoice Approval — [Vendor Name] — RM[InvoiceAmount]

STEP 6: Handle Outcomes in Both Branches

15. After each approval action, add Condition: Outcome = Approve
16. YES: Send approval notification email + add Excel row with Status = Approved
17. NO: Send rejection email with reason + add Excel row with Status = Rejected

Validation Checklist

Done?	Checklist Item
☐	Form has all required invoice fields
☐	Variable correctly captures Invoice Amount from form
☐	Condition correctly routes to dual or single approval based on amount
☐	Dual approval requires both approvers to respond
☐	Single approval responds when one approver decides
☐	Excel tracker updated with correct status and decision
☐	Email notifications sent for both Approved and Rejected outcomes
☐	Tested with one amount below RM5,000 AND one above



Run History shows Succeeded for both test runs

My Notes & Observations

What happens if one approver approves but the other rejects in 'Everyone must approve'?

How would you handle invoice approval timeout (no response after 3 days)?

LAB 8 — Employee Onboarding Workflow

Business Scenario

When a new employee joins DataTech Corp, HR must coordinate across 5 departments: send a welcome email, notify IT to set up accounts, notify Facilities for a workstation, create onboarding tasks in Planner, and get the manager's confirmation. This lab automates the entire onboarding coordination with one trigger.

Component	Configuration
Trigger	When an item is created in SharePoint Onboarding list
Action 1	Send welcome email to new employee
Action 2	Send IT setup email with equipment requirements
Action 3	Send HR notification email
Action 4	Post Teams message in HR channel
Action 5	Create Planner task: Workstation Setup
Action 6	Create Planner task: Welcome Kit Preparation
Action 7	Start approval — manager confirms onboarding checklist
Connector	SharePoint + Outlook + Teams + Planner + Approvals
Estimated Build Time	30–35 minutes

Pre-Lab Setup

18. In SharePoint, create list: New Employee Onboarding
19. Columns: Employee Name, Employee Email, Department, Job Title, Start Date, Manager Email, Equipment Required (Multi-line text), IT Access Needed (Multi-line text)
20. In Microsoft Planner, create a plan: Onboarding Tasks 2026

Building the Flow

STEP 1: Create Flow with SharePoint Trigger

21. Create Automated cloud flow: HR-NewEmployee-OnboardingFlow
22. Trigger: When an item is created (SharePoint — New Employee Onboarding list)

STEP 2: Add All Notification Actions

23. Action 1 — Welcome Email to Employee:
24. To: Employee Email (dynamic content)

25. Subject: Welcome to [Company Name], [Employee Name]!
26. Body: warm welcome message with Start Date, Job Title, and first day instructions
27. Action 2 — IT Setup Notification:
28. To: it@company.com
29. Subject: New Employee IT Setup Required — [Employee Name] — Start Date: [Start Date]
30. Body: include Equipment Required and IT Access Needed from SharePoint
31. Action 3 — HR Manager Notification:
32. To: hr@company.com
33. Subject: New Employee Joining — [Employee Name] | [Department]
34. Body: full employee details summary

STEP 3: Post Teams Message

35. + New Step > Post a message in a chat or channel
36. Post in: Channel — HR Team > General
37. Message: NEW EMPLOYEE ALERT: [Employee Name] is joining [Department] on [Start Date].
Please ensure onboarding preparations are complete.

STEP 4: Create Planner Tasks

38. + New Step > Create a task (Microsoft Planner)
39. Group Id: select your team
40. Plan Id: Onboarding Tasks 2026
41. Title: Setup Workstation for [Employee Name]
42. Due Date: Start Date (dynamic content)
43. Add second Create a task action:
44. Title: Prepare Welcome Kit for [Employee Name]

STEP 5: Manager Approval / Confirmation

45. + New Step > Start and wait for an approval
46. Title: Onboarding Checklist Confirmation — [Employee Name]
47. Assigned to: Manager Email (dynamic content)
48. Details: Please confirm that all onboarding preparations are in order for [Employee Name] starting [Start Date].
49. After approval: Send email to HR confirming manager acknowledgment

Validation Checklist

Done?	Checklist Item
☐	Trigger fires when new record added to SharePoint list
☐	Welcome email received by employee with correct details
☐	IT team receives equipment requirements notification
☐	HR manager receives new hire summary

☐	Teams channel shows new employee announcement
☐	Planner tasks created with correct titles and due dates
☐	Manager receives approval/confirmation request
☐	HR notified after manager confirms
☐	Run History shows all steps succeeded

My Notes & Observations

How many emails/notifications did a single SharePoint entry trigger?

What additional steps would you add for your organisation's onboarding process?

LAB 9 — AI Email Classification Flow

Business Scenario

TelecomPlus Bhd's customer service team receives 500+ emails weekly to support@telecomplus.com. Staff manually read each email and route it to the correct team. This lab uses AI Builder's text classification to automatically categorise incoming emails (Billing, Technical, Complaint, General) and route them to the appropriate team.

NOTE: AI Builder requires AI Builder credits (included in Power Automate Premium or Microsoft 365 E3/E5). Your trainer will advise if this feature is available in your tenant. If not available, you can complete Steps 1-2 and simulate the AI output using a Switch control with manual test values.

Component	Configuration
Trigger	When a new email arrives (shared mailbox or personal)
Action 1	Predict (AI Builder — Category Classification)
Action 2	Switch on AI prediction result
Billing Case	Forward to billing@company.com + create SharePoint ticket
Technical Case	Forward to tech@company.com + post Teams alert
Complaint Case	Forward to complaints@company.com + urgent Teams alert
Default Case	Forward to admin@company.com for manual review
Action 3	Apply email category label in Outlook
Estimated Build Time	25–30 minutes

Building the Flow

STEP 1: Create Flow with Email Trigger

50. Create Automated cloud flow: CS-AIEmailClassification-Flow
51. Trigger: When a new email arrives (V3)
52. Click Show advanced options > Subject Filter: leave blank (process all)
53. Only with Attachments: No

STEP 2: Add AI Builder Prediction

54. + New Step > Search: Predict
55. Select Predict (AI Builder)

- 56. Model: select Category classification (if available)
- 57. If not available, skip to Step 3 and use a static Switch for testing
- 58. Text: select Body (from email trigger dynamic content)

STEP 3: Add Switch Control

- 59. + New Step > Search: Switch
- 60. Select Switch (Control)
- 61. On: select predictionOutput?['results']?[0]?['label'] from dynamic content
- 62. OR for simulation: use a fixed value like 'Billing' to test one branch
- 63. Add Case for each category:

STEP 4: Configure Each Case

- 64. Case 1 — Billing:
- 65. Forward email to billing@company.com
- 66. Create SharePoint item in Customer Tickets list — Category: Billing
- 67. Case 2 — Technical:
- 68. Forward email to tech@company.com
- 69. Post Teams message in IT Support channel: New Technical Support Request
- 70. Case 3 — Complaint:
- 71. Forward email to complaints@company.com
- 72. Post URGENT Teams message in Customer Service channel
- 73. Send immediate acknowledgment email to customer
- 74. Default Case:
- 75. Forward email to admin@company.com
- 76. Create SharePoint item — Category: Unclassified / Needs Review

STEP 5: Apply Outlook Category Label

- 77. After the Switch, + New Step
- 78. Search: Categorize email (V2)
- 79. Message Id: Message Id from trigger dynamic content
- 80. Categories: select or type the AI classification result

Validation Checklist

Done?	Checklist Item
☐	Flow triggers when new email arrives
☐	AI Builder Predict action returns a classification result
☐	Switch control routes to correct branch based on category
☐	Billing emails forwarded to billing team
☐	Technical emails forwarded to IT team with Teams alert

☐	Complaint emails trigger urgent Teams notification
☐	Unclassified emails go to admin for manual review
☐	Customer receives acknowledgment email for complaints
☐	SharePoint ticket created for each classified email
☐	Run History shows Succeeded

My Notes & Observations

Which AI Builder model did you use? What was the prediction result for your test email?

What other business processes could benefit from AI-powered email classification?

CAPSTONE PROJECT — Build Your Own Automation

Your Mission

Using everything you have learned in the 9 labs, you will now design and build your own Power Automate flow to solve a real problem from YOUR workplace or the case studies provided. This is your chance to demonstrate mastery.

Project Planning Sheet

Flow Name (format: Dept-Process-Action):

Business Problem: What manual task are you automating?

Trigger: What event starts your flow?

Actions (list all steps your flow will perform):

Conditions / Logic: Any if/else decisions or loops?

Connectors Used (e.g. Outlook, SharePoint, Teams):

Variables Needed:

Error Handling: How will you handle failures?

Expected Business Benefit (time saved, errors reduced etc.):

Evaluation Rubric

Criteria & Marks	Description
Process Analysis — 25 marks	Clearly identified business problem, manual process documented, automation opportunity well-justified
Flow Design — 20 marks	Logical flow with correct trigger, actions, conditions; clear flow structure
Flow Construction — 30 marks	Working flow in Power Automate, correct connector use, dynamic content, error handling
Testing & Validation — 15 marks	Run History shows success, multiple scenarios tested, errors documented and resolved
Presentation — 10 marks	Clear explanation of solution, business value communicated, live demo or screenshots

Certification

Participants who score 60/100 or above will receive the Certificate of Completion for Microsoft Power Automate: Workflow Automation for Workplace Productivity. Issued by Dr. Fakhrul Syahmi | Data Consultant & Certified Trainer | mufasyahcons@gmail.com | © 2026

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